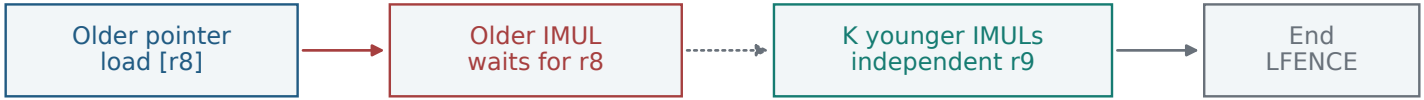
1 Workload and dependencies



Left to right is program order; solid red arrow is a true data dependency.

Control: XOR reads the older result (rAX) instead of ready r11; every block ends with LFENCE.

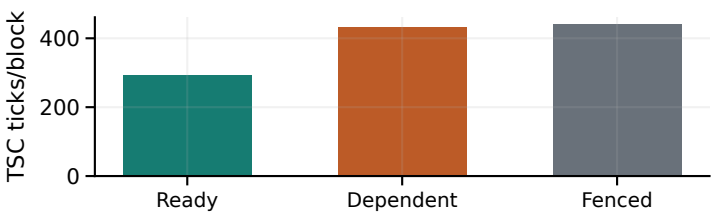
2 Controlled comparison

K=0 or 64 younger IMULs; independent / dependent / fenced; 16 KiB / 256 MiB.

3 Timed batch and normalization

I=100,000 fenced blocks; ticks/block = elapsed TSC / I.

4 Measured response



5 Inference and control

The younger chain is ready while an older IMUL waits for the pointer load. Making it depend on that result, or fencing before it, raises cold K=64 time. A common end fence blocks overlap with the next iteration.